

CS2601 Linear and Convex Optimization

Homework 2

Due: 2023.10.17

1. Let $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ be an affine function from $\mathbb{R}^n$ to $\mathbb{R}^m$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Show that if $C \subset \mathbb{R}^m$ is convex, so is its inverse image $f^{-1}(C) \triangleq \{\mathbf{x} : f(\mathbf{x}) \in C\}$.
2. Suppose C_1, C_2 are two nonempty convex sets with $C_1 \cap C_2 = \emptyset$. Show $C = C_1 - C_2 = \{\mathbf{x}_1 - \mathbf{x}_2 : \mathbf{x}_1 \in C_1, \mathbf{x}_2 \in C_2\}$ is a nonempty convex set and $\mathbf{0} \notin C$.
3. Suppose C is a convex set.
 - (a). Show that its interior $\text{int } C$ is convex.
 - (b). Show that its closure $\bar{C}$ is convex.
4. Show by induction on m that if $C \subset \mathbb{R}^n$ is convex and $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in C$, then any convex combination $\sum_{i=1}^m \theta_i \mathbf{x}_i \in C$. (We have sketched the proof. Please fill in the details.)
5. Let $\mathbf{x}_0, \dots, \mathbf{x}_K$ be distinct points in $\mathbb{R}^n$. Let V be the set of points that are closer in Euclidean distance to $\mathbf{x}_0$ than $\mathbf{x}_1, \dots, \mathbf{x}_K$, called the **Voronoi region** around $\mathbf{x}_0$ with respect to $\mathbf{x}_1, \dots, \mathbf{x}_K$, i.e.

$$V = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\|_2 \leq \|\mathbf{x} - \mathbf{x}_i\|_2, i = 1, 2, \dots, K\}.$$

Show that V is a polyhedron by identifying $\mathbf{A}$ and $\mathbf{b}$ such that $V = \{\mathbf{x} : \mathbf{A}\mathbf{x} \leq \mathbf{b}\}$. (You don't have to draw it in the submission, but try to visualize V for $\mathbb{R}^2$.)